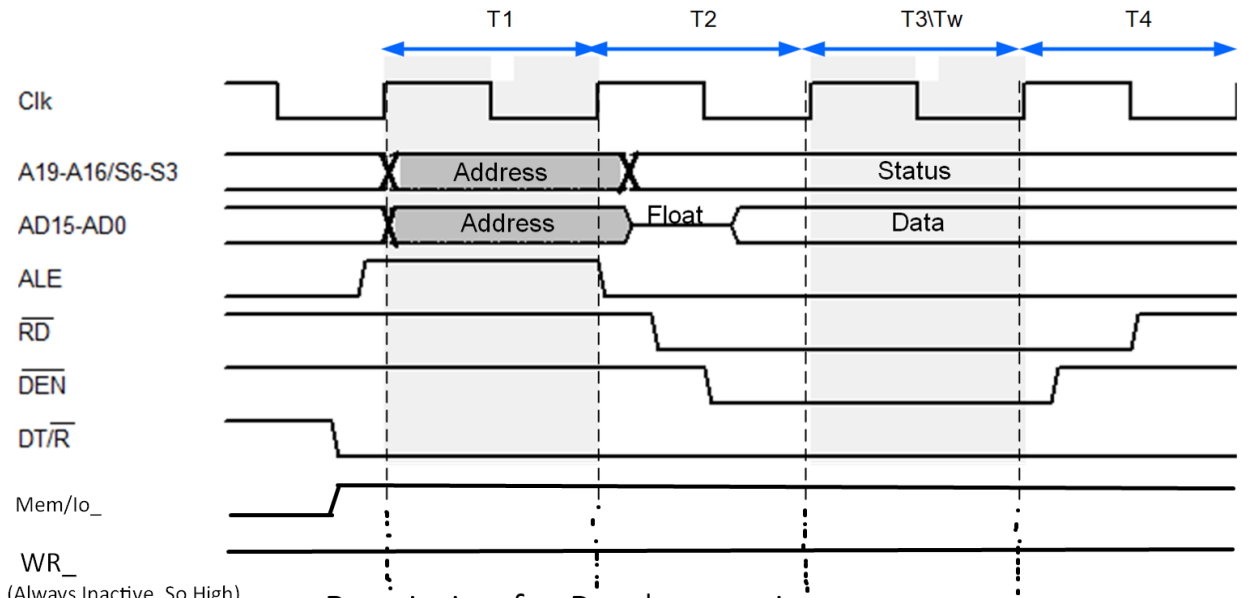
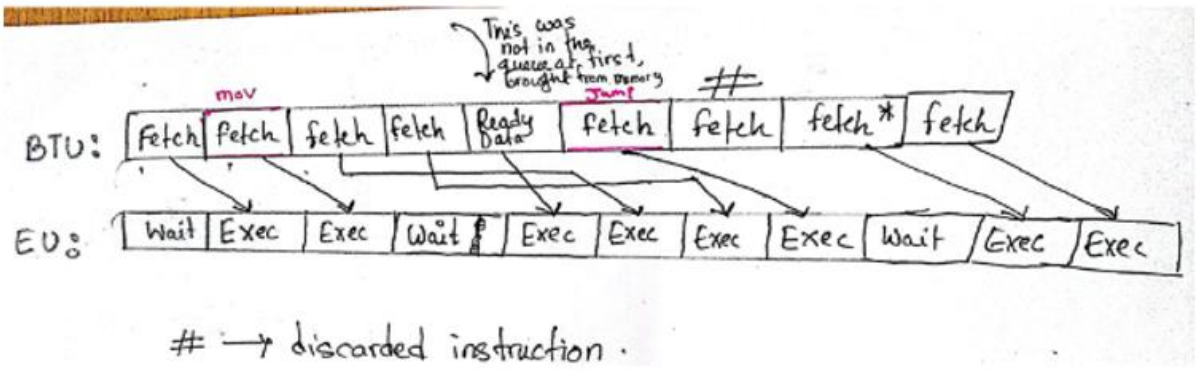
2 (a)



Bus timing for Read operation

Activate Windows

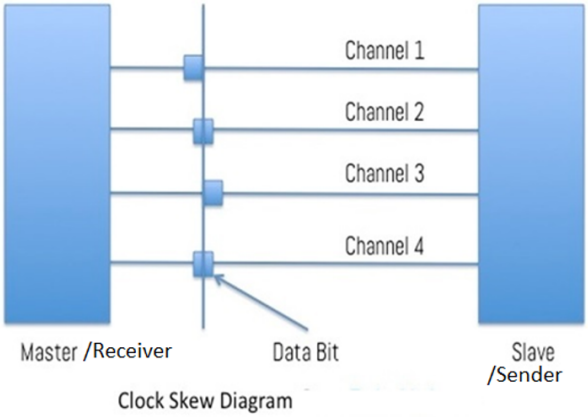
(b)



(C)

Terminology

Clock Skew: In a parallel circuit, clock skew phenomenon is observed, which is the time difference in the arrival of two bits sequentially in adjacent registers. There is a time lag in the data bits through different channels of the same bus. Clock skew is inevitable due to differences in physical conditions of the channels, like temperature, resistance, path length, etc



Clock Skew Diagram

For example, around 5 people are firing at the same time, there is bound to be a time difference in the arrival of the bullet from the first shooter and that from the second shooter and so on. This time difference is what we call clock skew.

Activate Windows
Go to Settings to activate Windows.

3(a)

```
LDA  CH
ADD  DH
ADD  DH
SUB  FH
OUT
HLT
```

3(b)

0H : 1100 1100
1H : 0100 1101
2H : 0100 1101
3H : 0011 1111
4H : 0001 xxxx
5H : 1000 xxxx

3(C)

Reg A	Reg B
27 or 1B H	0 / 0H
52 / 34 H	25 / 19 H
77 / 4D H	25 / 19 H
42 / 2A H	35 / 23 H